

Unit II Combinatorics

Course Objective – Cob2: To familiarize students in combinatorial problems and its applications.

Course Outcome – CO2: To use counting principles and mathematical induction in solving combinatorial problems.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BT L*
1.	State the principle of induction. Suppose $P(n)$ is a mathematical statement imposed on every positive integer n, To Prove: $P(n)$ is true $\forall n \in \mathbb{Z}^+$ Basic Step: To Prove $P(1)$ is true Induction Step: Assume that $P(k)$ is true, where k is some positive integer To Prove: $P(k + 1)$ is true. Hence by Induction Principle $P(n)$ is true $\forall n \in \mathbb{Z}^+$.	K1
2.	Use mathematical induction to show that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ Given that $P(n) : 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ To Prove: $P(n)$ is true $\forall n \in \mathbb{Z}^+$ Basic Step: To Prove $P(1)$ is true Proof: Substitute $n = 1$ in $P(n)$ LHS of $P(n) \Rightarrow 1$ RHS of $P(n) \Rightarrow \frac{1(2)}{2} = 1$ Hence $P(1)$ is true Induction Step: Assume that $P(k)$ is true, where k is some positive integer (i.e) $1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$(1) To Prove: $P(k + 1)$ is true. $1 + 2 + 3 + \dots + (k + 1) = \frac{(k + 1)[(k + 1) + 1]}{2}$ Proof: Substitute $n = k + 1$ in $P(n)$ LHS of $P(k + 1)$ $\Rightarrow 1 + 2 + 3 + \dots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1) \quad (\text{since by (1)})$ $= (k + 1) \left[\frac{k}{2} + 1 \right]$ $= \frac{(k+1)(k+2)}{2}$ $= \frac{(k+1)[(k+1)+1]}{2}$ $= \text{RHS of } P(k + 1)$ Hence $P(k + 1)$ is true Hence by Induction Principle $P(n)$ is true $\forall n \in \mathbb{Z}^+$.	K2

3.	State the pigeonhole principle. If n pigeons are accommodated in m pigeon-holes and $n > m$, then at least one pigeon-holes will contain 2 or more pigeons.	K1
4.	If seven colours are used paint to 40 bikes, then show that at least 6 bikes will be the same colour. Let $n = 40$ bikes be pigeons, Let $m = 7$ colours be pigeon-holes, By Generalised Pigeon Hole Principle, At least $k = \left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons will be in the same hole At least $k = \left\lfloor \frac{40-1}{7} \right\rfloor + 1$ $= 5 + 1 = 6$ bikes will be in same colours	K2
5.	What is the minimum number of students required in a Number Theory class to be sure that at least seven will receive the same grade if there are six possible grades O, A+, A, B+, B and C? Let n be minimum number of students (pigeons) required in the class. Let $m = 6$ grades be pigeon-holes, And Given that at least seven will receive the same grade By Generalised Pigeon Hole Principle, At least $k = \left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons will be in the same hole $7 = \left\lfloor \frac{n-1}{6} \right\rfloor + 1$ students (pigeons) will receive the same grade $\Rightarrow 7 = \left(\frac{n-1}{6} \right) + 1 \Rightarrow 42 = (n-1) + 6$ $\Rightarrow n = 37$ is minimum number of students required in a Number Theory class to be sure that at least seven will receive the same grade	K2
6.	How many 3-letter words can be formed from the letters of the word NUMBER without repetition and with repetition. Given that 6-letter word "NUMBER" Permutation of 6 letters taken } $6P_3 = \frac{6!}{(6-3)!} = \frac{6!}{3!} = 6 \times 5 \times 4 = 120$ words 3 at time without repetition is Permutation of 6 letters taken } $6^3 = 6 \times 6 \times 6 = 216$ words 3 at time with repetition is	K2
7.	How many different words are there in the following words? (ii) MALAYALAM (ii) UNUSUAL (iii) HAPPINESS (i) Permutation of 9 letters in which M repeats } $\frac{9!}{4! \times 2! \times 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5}{2! \times 2!}$ 2 times, A repeats 4 times, L repeats 2 times is } $= 9 \times 4 \times 7 \times 3 \times 5$ $= 3780$ words (ii) Permutation of 7 letters in } $\frac{7!}{3!} = 7 \times 6 \times 5 \times 4 = 840$ words which U repeats 3 times is } (iii) Permutation of 9 letters in which } $\frac{9!}{2! \times 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{2!}$ P repeats 2 times, S repeats 2 times is } $= 9 \times 8 \times 7 \times 6 \times 5 \times 2 \times 3$ $= 90720$ words	

8.	How many bit strings of length ten contain (i) exactly four 1's, (ii) at least four 1's? (i) The number of strings of length ten that contain exactly four 1's is $10C_4 = 210$ (ii) The number of strings of length ten that contain at least four 1's $= 10C_4 + 10C_5 + 10C_6 + 10C_7 + 10C_8 + 10C_9 + 10C_{10} = 848$	K2
9.	Find the recurrence relation for the Fibonacci sequence. The recurrence relation for the Fibonacci sequence is given by $F(n) = F(n-1) + F(n-2), \quad n \geq 2$ with initial values $F(0) = 0, F(1) = 1$ (OR) $F(n+2) = F(n+1) + F(n), \quad n \geq 0$ with initial values $F(0) = 0, F(1) = 1$	K2
10.	Find the recurrence relation of the sequence $s(n) = a^n; n \geq 1$ Given that $S(n) = a^n \dots \dots \dots (1)$ $S(n-1) = a^{n-1} \dots \dots \dots (2)$ $\frac{(1)}{(2)} \Rightarrow \frac{S(n)}{S(n-1)} = \frac{a^n}{a^{n-1}} = \frac{a^n}{a^n a^{-1}} = a$ $\Rightarrow S(n) = a S(n-1)$ $\Rightarrow S(n) - a S(n-1) = 0, \quad n \geq 2, \text{ with } S(1) = a$	K2
11.	Find the generating function for the sequence $S(n) = n, n = 1, 2, 3 \dots$ W.K.T the Generating function of the sequence $\{a_0, a_1, a_2, a_3, \dots a_n, \dots\}$ is given by $G(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots a_nx^n + \dots$ Now Given that $Seq\{S(n)\}_{n=0}^\infty = Seq\{n+1\}_{n=0}^\infty = \{1, 2, 3, 4, 5, \dots (n+1) \dots\}$ Then the Generating function of the sequence is given by $G(x) = 1 + 2x + 3x^2 + 4x^3 + \dots$ $G(x) = (1-x)^{-2} = \frac{1}{(1-x)^2}$	K2
12.	Find the generating function for the sequence $1, a, a^2, a^3 \dots$ W.K.T the Generating function of the sequence $\{a_0, a_1, a_2, a_3, \dots a_n, \dots\}$ is given by $G(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots a_nx^n + \dots$ Now the Given sequence is $\{1, a, a^2, a^3 \dots\}$ Then the Generating function of the sequence is given by $G(x) = 1 + ax + a^2x^2 + a^3x^3 + \dots$ $G(x) = 1 + ax + (ax)^2 + (ax)^3 + \dots$ $G(x) = (1-ax)^{-1} = \frac{1}{1-ax}$	K2

Part B

S.No.	Question	BTL *
1.	Using mathematical induction show that (i) $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ (ii) $\sum_{r=0}^n 3^r = \frac{3^{n+1}-1}{2}$ (iii) $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n	K4
2.	(i) Prove that in any group of 6-people at least 3 must be mutual friends or at least 3 must be mutual strangers. (ii) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen? (iii) Let b be an integer ≥ 2 . Suppose $b + 1$ integers are randomly selected. Prove that the difference of two of them is divisible by b .	K4
3.	(i) How many positive integers n can be formed using the digits 3,4,4,5,5,6,7 if n has to exceed 5000000? (ii) How many numbers lying between 100 and 1000 can be formed with the digits 0, 1, 2, 3, 4, 5, 6. If the repetition of the digits is not allowed. (iii) Find the number of arrangement of the letters of the word ENGINEERING. In how many of these arrangement (a) Do the word start with R. (b) Do all the vowels always occur together. (c) Do the vowels never occur together.	K4
4.	(i) What is the number of ways of choosing 4 cards from a pack of 52 playing cards? In how many of these (a) Four cards are of the same suit. (b) Four cards belong to four different suits. (c) Four cards are of face cards. (d) two are red and two are black cards. (e) Four cards are of the same colour. (ii) A box contains six white balls and five red balls. Find the number of ways four balls can be drawn from the box if (1) they can be of any colour (2) two must be white and two red, (3) they are of the same colour. (iii) A committee of 7 has to be formed from 8 boys and 5 girls. How many ways can this be done when the committee consists of (a) exactly 3 girls (b) at least 3 girls (c) at most 3 girls	K4
5.	(i) Find the generating function of Fibonacci sequence. (ii) Using generating function solve $y_{n+2} - 5y_{n+1} + 6y_n = 0$ with $y_0 = 1$ & $y_1 = 1$ (iii) Using method of generating function to solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2} + 4^n, n \geq 2$ given that $a_0 = 2$ & $a_1 = 8$	K4

6.	(i) A total of 1232 students have taken a course in Spanish, 879 have taken a course in French and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French and Russian, how many students have taken a course in all three languages? (ii) There are 2500 students in a college, of these 1700 have taken a course in C, 1000 have taken a course Pascal and 550 have taken a course in Networking. Further 750 have taken courses in both C and Pascal. 400 have taken courses in both C and Networking and 275 have taken courses in both Pascal and Networking. If 200 of these students have taken courses in C, Pascal and Networking. (A). How many of these 2500 students have taken a course in any one of these three courses C, Pascal and Networking? (B). How many of these 2500 students have not taken a course in any one of these three courses C, Pascal and Networking?	K4
----	---	----

A standard deck of 52 cards has 13 kinds of cards, with four cards of each of kind, one in each of the four suits, hearts, diamonds, spades, and clubs.